

# Python Data Structures Cheat Sheet

## List

| Package/Method   | Description                                                                                                                                                                                                                                                 | Code Example                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| append()         | The `append()` method is used to add an element to the end of a list.                                                                                                                                                                                       | Syntax:<br><br>list_name.append(element)<br><br>Example:<br><br>fruits = ["apple", "banana", "orange"]<br>fruits.append("mango") print(fruits)                                                                                                                                                                                                                                          |
| copy()           | The `copy()` method is used to create a shallow copy of a list.                                                                                                                                                                                             | Example 1:<br><br>my_list = [1, 2, 3, 4, 5]<br>new_list = my_list.copy() print(new_list)<br># Output: [1, 2, 3, 4, 5]                                                                                                                                                                                                                                                                   |
| count()          | The `count()` method is used to count the number of occurrences of a specific element in a list in Python.                                                                                                                                                  | Example:<br><br>my_list = [1, 2, 2, 3, 4, 2, 5, 2]<br>count = my_list.count(2) print(count)<br># Output: 4                                                                                                                                                                                                                                                                              |
| Creating a list  | A list is a built-in data type that represents an ordered and mutable collection of elements. Lists are enclosed in square brackets [] and elements are separated by commas.                                                                                | Example:<br><br>fruits = ["apple", "banana", "orange", "mango"]                                                                                                                                                                                                                                                                                                                         |
| del              | The `del` statement is used to remove an element from list. `del` statement removes the element at the specified index.                                                                                                                                     | Example:<br><br>my_list = [10, 20, 30, 40, 50]<br>del my_list[2] # Removes the element at index 2 print(my_list)<br># Output: [10, 20, 40, 50]                                                                                                                                                                                                                                          |
| extend()         | The `extend()` method is used to add multiple elements to a list. It takes an iterable (such as another list, tuple, or string) and appends each element of the iterable to the original list.                                                              | Syntax:<br><br>list_name.extend(iterable)<br><br>Example:<br><br>fruits = ["apple", "banana", "orange"]<br>more_fruits = ["mango", "grape"]<br>fruits.extend(more_fruits)<br>print(fruits)                                                                                                                                                                                              |
| Indexing         | Indexing in a list allows you to access individual elements by their position. In Python, indexing starts from 0 for the first element and goes up to `length_of_list - 1`.                                                                                 | Example:<br><br>my_list = [10, 20, 30, 40, 50]<br>print(my_list[0])<br># Output: 10 (accessing the first element)<br>print(my_list[-1])<br># Output: 50 (accessing the last element using negative indexing)                                                                                                                                                                            |
| insert()         | The `insert()` method is used to insert an element.                                                                                                                                                                                                         | Syntax:<br><br>list_name.insert(index, element)<br><br>Example:<br><br>my_list = [1, 2, 3, 4, 5]<br>my_list.insert(2, 6)<br>print(my_list)                                                                                                                                                                                                                                              |
| Modifying a list | You can use indexing to modify or assign new values to specific elements in the list.                                                                                                                                                                       | Example:<br><br>my_list = [10, 20, 30, 40, 50]<br>my_list[1] = 25 # Modifying the second element<br>print(my_list)<br># Output: [10, 25, 30, 40, 50]                                                                                                                                                                                                                                    |
| pop()            | `pop()` method is another way to remove an element from a list in Python. It removes and returns the element at the specified index. If you don't provide an index to the `pop()` method, it will remove and return the last element of the list by default | Example 1:<br><br>my_list = [10, 20, 30, 40, 50]<br>removed_element = my_list.pop(2) # Removes and returns the element at index 2<br>print(removed_element)<br># Output: 30<br>print(my_list)<br># Output: [10, 20, 40, 50]<br><br>Example 2:<br><br>my_list = [10, 20, 30, 40, 50]<br>removed_element = my_list.pop() # Removes and returns the last element<br>print(removed_element) |

|           |                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
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|           |                                                                                                                                                                                                   | <pre># Output: 50 print(my_list) # Output: [10, 20, 30, 40]</pre>                                                                                                                                                                                                                                                                                                                                          |
| remove()  | To remove an element from a list. The `remove()` method removes the first occurrence of the specified value.                                                                                      | <p>Example:</p> <pre>my_list = [10, 20, 30, 40, 50] my_list.remove(30) # Removes the element 30 print(my_list) # Output: [10, 20, 40, 50]</pre>                                                                                                                                                                                                                                                            |
| reverse() | The `reverse()` method is used to reverse the order of elements in a list                                                                                                                         | <p>Example 1:</p> <pre>my_list = [1, 2, 3, 4, 5] my_list.reverse() print(my_list) # Output: [5, 4, 3, 2, 1]</pre>                                                                                                                                                                                                                                                                                          |
| Slicing   | You can use slicing to access a range of elements from a list.                                                                                                                                    | <p>Syntax:</p> <pre>list_name[start:end:step]</pre> <p>Example:</p> <pre>my_list = [1, 2, 3, 4, 5] print(my_list[1:4]) # Output: [2, 3, 4] (elements from index 1 to 3) print(my_list[:3]) # Output: [1, 2, 3] (elements from the beginning up to index 2) print(my_list[2:]) # Output: [3, 4, 5] (elements from index 2 to the end) print(my_list[::-2]) # Output: [1, 3, 5] (every second element)</pre> |
| sort()    | The `sort()` method is used to sort the elements of a list in ascending order. If you want to sort the list in descending order, you can pass the `reverse=True` argument to the `sort()` method. | <p>Example 1:</p> <pre>my_list = [5, 2, 8, 1, 9] my_list.sort() print(my_list) # Output: [1, 2, 5, 8, 9]</pre> <p>Example 2:</p> <pre>my_list = [5, 2, 8, 1, 9] my_list.sort(reverse=True) print(my_list) # Output: [9, 8, 5, 2, 1]</pre>                                                                                                                                                                  |

## Tuple

| Package/Method  | Description                                                                                                                                                                  | Code Example                                                                                                                                                                                                         |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| count()         | The count() method for a tuple is used to count how many times a specified element appears in the tuple.                                                                     | <p>Syntax:</p> <pre>tuple.count(value)</pre> <p>Example:</p> <pre>fruits = ("apple", "banana", "apple", "orange") print(fruits.count("apple")) #Counts the number of times apple is found in tuple. #Output: 2</pre> |
| index()         | The index() method in a tuple is used to find the first occurrence of a specified value and returns its position (index). If the value is not found, it raises a ValueError. | <p>Syntax:</p> <pre>tuple.index(value)</pre> <p>Example:</p> <pre>fruits = ("apple", "banana", "orange") print(fruits[1]) #Returns the value at which apple is present. #Output: banana</pre>                        |
| sum()           | The sum() function in Python can be used to calculate the sum of all elements in a tuple, provided that the elements are numeric (integers or floats).                       | <p>Syntax:</p> <pre>sum(tuple)</pre> <p>Example:</p> <pre>numbers = (10, 20, 5, 30) print(sum(numbers)) #Output: 65</pre>                                                                                            |
| min() and max() | Find the smallest (min()) or largest (max()) element in a tuple.                                                                                                             | <p>Example:</p> <pre>numbers = (10, 20, 5, 30) print(min(numbers)) #Output: 5 print(max(numbers)) #Output: 30</pre>                                                                                                  |
| len()           | Get the number of elements in the tuple using len().                                                                                                                         | <p>Syntax:</p> <pre>len(tuple)</pre>                                                                                                                                                                                 |

Example:

```
fruits = ("apple", "banana", "orange")  
print(len(fruits)) #Returns length of the tuple.  
#Output: 3
```



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